

Week 2 Plan Explanation – Factor Foundations (Value & Momentum)

Week Goal

Implement two classic factors from scratch, form sorted portfolios, and evaluate their performance.

By the end of the week, the intern can define value and momentum signals without timing leakage, form monthly quintile portfolios, and communicate factor evidence with appropriate caveats.

Task 1: Factor Concepts & Value Factor

This task introduces factor logic through the economic intuition and implementable definition of a value signal.

Detailed Work

- Read about Fama–French factors and their economic intuition
- Value factor definition: Book–to–Market ratio (B/M) or Earnings Yield (E/P)
- If fundamental data is unavailable, use a simplified proxy (e.g., inverse of trailing 12–month return as contrarian value)

Concrete Output

- Compute a value score for each stock at each rebalancing date
- Rank stocks cross–sectionally by value score

Quality Check

The signal definition, data availability date, ranking direction, and fallback proxy must be explicit enough for another analyst to reproduce.

Task 2: Momentum Factor

This task constructs the classic 12–1 momentum signal while enforcing the timing convention that avoids short–term reversal and look–ahead bias.

Detailed Work

- Momentum definition: 12–1 month momentum (cumulative return from $t-12$ to $t-1$, skipping the most recent month)
- Why skip the most recent month: short–term reversal effect

Concrete Output

- Compute 12–1 momentum signal for all stocks
- Verify signal calculation: spot–check a few stocks manually
- Plot the distribution of momentum scores at a given date

Quality Check

The most recent month must be excluded, signals must be lagged before portfolio formation, and several observations must match manual calculations.

Task 3: Portfolio Sorting & Quintile Portfolios

This task converts cross–sectional signals into monthly rebalanced quintile portfolios whose next–period returns can be evaluated.

Detailed Work

- Cross–sectional sorting methodology:
 - At each rebalance date, rank all stocks by factor score
 - Divide into quintiles (Q1 = lowest, Q5 = highest)
 - Compute equal–weight returns for each quintile in the next period
- Rebalancing frequency: monthly

Concrete Output

- Build a reusable `form_quintile_portfolios(df, signal_col, ret_col)` function in `src/factor_utils.py`
- Apply to both value and momentum factors
- Compute monthly returns for Q1 through Q5

Quality Check

Membership must be determined using information available at the rebalance date, and returns must come strictly from the following holding period.

Task 4: Factor Performance Analysis

This task evaluates whether the sorted portfolios display a coherent return spread and whether that spread survives basic risk analysis.

Detailed Work

- Plot cumulative returns for each quintile (Q1–Q5)
- Compute long–short spread: Q5 minus Q1 (or Q1 minus Q5 depending on factor direction)
- Performance metrics: annualized return, volatility, Sharpe ratio, max drawdown

Concrete Output

- Create summary table of quintile portfolio statistics
- Plot year–by–year factor returns – are there periods of reversal?
- Compare value and momentum: correlation of their long–short returns

Quality Check

All portfolios must use identical rebalancing and annualization conventions, and the long–short direction must agree with the stated factor hypothesis.

Task 5: Factor Memo & Discussion

This task turns the calculations into a concise research argument that distinguishes evidence, assumptions, and limitations.

Detailed Work

- Factor–neutral vs market–neutral: what's the difference?
- Discuss: why might value and momentum be negatively correlated?
- Limitations: data snooping, small sample, survivorship bias

Concrete Output

- Write a 1–2 page factor memo covering methodology, results, and caveats
- Friday demo: present quintile charts and key findings to mentor

Quality Check

The memo must separate findings from interpretation, discuss risk adjustment and survivorship bias, and avoid claiming causality from a backtest.

Week Deliverables

- `notebooks/week2/factor_value_momentum.ipynb`
- `src/factor_utils.py` – factor computation and quintile formation
- Factor memo (1–2 pages): methodology, results, assumptions, pitfalls

Recommended Reading

- Fama & French (1993), "Common risk factors in the returns on stocks and bonds"
- Jegadeesh & Titman (1993), "Returns to buying winners and selling losers"
- *Expected Returns* (Ilmanen), Chapters on Value and Momentum

Common Pitfalls

- Forgetting to lag the signal (using same-period data = look-ahead bias)
- Not handling delisted stocks (survivorship bias inflates returns)
- Comparing quintile returns without risk adjustment

What Success Looks Like

The intern can define value and momentum signals without timing leakage, form monthly quintile portfolios, and communicate factor evidence with appropriate caveats.